



B.E. Project Guidelines for Students

Academic Year:2026-27

1. Objective

The B.E. Project is the culmination of the undergraduate engineering program and aims to provide students with an opportunity to apply theoretical knowledge to solve real-world engineering problems through innovation, research, design, implementation, testing, and documentation. The project should demonstrate technical competency, creativity, teamwork, communication skills, research aptitude, and professional ethics.

2. Project Group Formation

- Each project group shall consist of 3–4 students (as per SPPU/Institute guidelines).
- Every member must actively participate throughout the project duration.
- Any change in project group or project title after approval requires prior approval from the Project Coordinator.

3. Project Topic Selection

Students are encouraged to select projects based on:

- Real-world industrial problems
- Research-oriented problems
- Societal applications
- Product development
- Start-up ideas
- Smart India Hackathon (SIH) problem statements
- Government initiatives
- Industry-sponsored projects

Projects should preferably be based on emerging technologies such as:

- Artificial Intelligence
- Machine Learning
- Deep Learning
- Data Science
- Cyber Security
- Cloud Computing
- Internet of Things (IoT)
- Blockchain
- Computer Vision
- Natural Language Processing



- Robotics
- AR/VR
- Big Data Analytics
- Software Engineering
- Mobile and Web Technologies

Students should avoid duplicate, copied, or ready-made projects.

4. Project Guide

- Every project group will be allotted a faculty guide.
- Students must meet the guide regularly (minimum once every week).
- Weekly progress shall be recorded in the Project Work Diary and Minutes of Meeting.
- Students must implement the suggestions given by the guide before the next review.

5. Synopsis Submission

Each group shall submit a maximum of three project proposals to the guide.

The synopsis should include:

- Problem Statement
- Motivation
- Objectives
- Literature Survey
- Research Gap
- Scope
- Methodology
- Proposed Algorithms
- Expected Outcomes

- Hardware and Software Requirements
- References

The final project topic will be approved by the Project Review Committee (PRC).

6. Project Reviews

Project work will be monitored through Eight (08) Project Reviews conducted during the academic year.

Semester I

Review I

- Problem Statement
- Motivation
- Objectives
- Literature Survey



Review II

- Feasibility Study
- Project Scope
- Technology Selection

Review III

- Requirement Analysis
- Software Requirement Specification (SRS)

Review IV

- System Design
- UML Diagrams
- Database Design
- Architecture Design

Semester II

Review V

- Model Refinement
- Algorithm Development

Review VI

- Coding and Implementation

Review VII

- Validation
- Testing
- Performance Evaluation
- Result Analysis

Review VIII

- Report Writing
- Documentation
- Final Demonstration Readiness

Important

- Students must attend all eight project reviews.
- Review schedules shall be conducted as per the Department Project Academic Calendar.
- PPT presentation, updated report, source code, work diary, and demonstration (where applicable) are compulsory during every review.



- Failure to appear for any review without valid permission may result in loss of internal marks.

7. Term Work Assessment

The Term Work (TW) marks for Project Stage-I and Project Stage-II shall be awarded based on:

- Performance in all Eight Project Reviews
- Regularity of meetings with the guide
- Progress of implementation
- Documentation quality
- Technical knowledge
- Presentation skills
- Demonstration
- Viva-Voce
- Teamwork
- Attendance during reviews

Continuous assessment will be carried out throughout the academic year.

8. Research Publication (Mandatory)

To promote research culture, every project group must publish TWO research papers based on their project.

Paper 1 (Mandatory) : Review Paper

The review paper should include:

- Comprehensive Literature Survey
- Existing Techniques
- Comparative Analysis
- Research Gap
- Future Scope

Paper 2 (Mandatory) : Research Implementation/Result Paper

The second paper should include:

- Proposed Methodology
- Implementation
- Experimental Results
- Performance Analysis
- Conclusion



Both papers should preferably be submitted to or published in:

- IEEE Conferences
- IEEE Journals
- Scopus Indexed Conferences
- Scopus Indexed Journals

Proper plagiarism guidelines and publication ethics must be followed.

9. Intellectual Property Rights (Highly Appreciated)

Students are strongly encouraged to protect their innovative work through:

- Copyright Registration
- Patent Filing
- Design Registration (where applicable)
- Software Registration
- Book Chapter Publication based on the project

Such achievements will be considered as value addition during project evaluation and departmental recognition.

10. Participation in Technical Competitions (Highly Recommended)

Every project group is encouraged to participate in:

- Smart India Hackathon (SIH)
- KPIT Sparkle
- IEEE Project Competitions
- National Level Project Competitions
- State Level Project Competitions
- Innovation Challenges
- Hackathons
- Ideathons
- Startup Competitions
- Research Symposiums

Participation certificates and awards should be submitted to the department.

11. Documentation

Students shall maintain the following documents throughout the project:

- Approved Synopsis



- Weekly Project Work Diary containing Minutes Of Meeting(MOM)
- WorkBook
- Literature Survey
- Software Requirement Specification (SRS)
- UML Diagrams
- Design Documents
- Algorithms
- Source Code
- Data Sets (if applicable)
- Test Cases
- Test Reports
- User Manual
- Installation Manual
- Stage I Project Report
- Stage II Project Report (Blackbook)
- Publication Certificates

12. Coding Standards

Students should follow:

- Modular Programming
- Standard Coding Practices
- Meaningful Naming Conventions
- Proper Comments
- Exception Handling
- Secure Coding Practices

13. Testing

The following testing activities should be performed wherever applicable:

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing (UAT)
- Performance Testing
- Security Testing

Testing reports shall form part of the final project report.

14. Final Project Report

The final report shall contain:



- Title Page
- Certificate
- Acknowledgement
- Abstract
- Table of Contents
- List of Figures
- List of Tables
- Introduction
- Literature Survey
- Requirement Analysis
- System Design
- Algorithms
- Implementation
- Testing
- Results and Discussion
- Conclusion
- Future Scope
- References
- Research Papers
- Copyright/Patent Details (if applicable)
- Book Chapter Details(if applicable)
- Project competition participation certificates
- Plagiarism Report
- Group photo including all team members & guide

15. Academic Integrity

Students shall:

- Avoid plagiarism in reports, research papers, and source code.
- Properly acknowledge all references.
- Use AI tools ethically and responsibly.
- Maintain academic honesty and professional ethics.

Violation of academic integrity may lead to disciplinary action.

16. Evaluation Criteria

The project evaluation shall be based on:

- Technical Knowledge
- Innovation
- Problem Solving
- Literature Survey



- Requirement Analysis
- System Design
- Algorithm Development
- Coding & Implementation
- Testing & Validation
- Documentation Quality
- Research Publications
- Presentation Skills
- Demonstration
- Viva-Voce
- Teamwork
- Professional Ethics

17. Final Deliverables

Each project group shall submit:

- Approved Synopsis
- Eight Review Presentations
- Weekly Work Diary containing Minutes of Meeting
- Workbook
- Source Code
- Final Project Report Stage I & Stage II (2 Hard copies & Soft Copy)
- Two Research Papers
- Copyright/Patent/Book Chapter Details (if applicable)
- Competition/Hackathon Participation Certificates (if any)
- Any other documents as instructed by the department.

18. General Instructions

- Attend all scheduled reviews and meetings.
- Follow the departmental Project Academic Calendar strictly.
- Complete assigned tasks before every review.
- Maintain backups of all project files.
- Every member should be able to explain every module of the project.
- Maintain professional conduct during reviews and examinations.
- Continuous progress is expected throughout the academic year. Last-minute implementation or report preparation is strongly discouraged.



Student Declaration

We have read and understood the above guidelines and agree to comply with all departmental rules and regulations throughout the B.E. Project.

Group No.: _____

Roll No	Name of Student	Signature

Date : _____

Prof. Shital Patil
Project Coordinator

Dr. Amrin Sheikh
HOD